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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

ACH202T – ANALYTICAL CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Sublimation phase transition is _____
   a) Solid-gas b) Solid-liquid c) Liquid-gas d) Gas-liquid

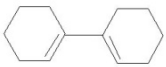
2. The temperature in steam distillation process
   a) Heigh temperature b) Low temperature
   c) Isolate temperature d) Cold condition

3. Using TLC method _____ can be separated
   a) Solid-liquid mixture b) Liquid-liquid
   c) Non volatile d) Volatile mixture

4. Formula of sodium zeolite is _____
   a) $NaAlsi_3o_3$ b) $NaAlsi_2o_6$ c) $Nasi_3o_6$ d) Nao_2si_4

5. In column chromatography _____is used as a mobile phase
   a) Solid b) Liquid c) Gas d) Solid-liquid

6. The principle of electron transition spectroscopy involves
   a) IR b) Raman c) UV d) Other

7. Calulate the λmax value of 

   a) 230mμ b) 212mμ c) 235mμ d) 232 mμ

8. Mercury electrode is otherwise called
   a) DME b) SMDE c) Cationic d) Anionic

9. One of the applications of polorimetry to find_____
   a) Optically active b) Optically inactive
   c) Polar compound d) Non polar

10. Desalination process uses _____
    a) Sea water b) Hard water c) Soft water d) Saline water

II. Fill up the blanks

11. An example for Inorganic precipitating agent is _____

12. _____ method is used to purify a solid organic compound.

13. Current voltage curve corresponds to _____

14. Beer lambert law gives a relationship between _____&_____

15. The molecular formula of EDTA is _____

III. Match the following

16. Solid particles	-	Wave length & frequency
17. Polarography	-	Non volatile mixture
18. λmax	-	Polarisable electrode
19. Desalination	-	Crystallization
20. TLC	-	Woodward Fischer

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Write a short note on specific and selective precipitants.

22. Define steam distillation.

23. Give various purification methods for solid organic compound?

24. What is crystallization?

25. State Ilkovic equation.

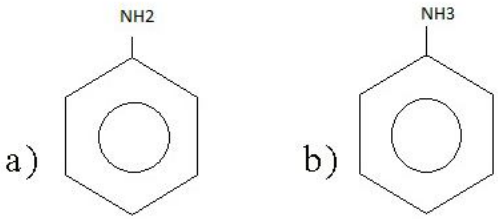
26. Write the principle of amperometry.

27. Give any four applications of polarimetry?

28. Define with examples: chromophore.

29. Define Auxochrome.

30. Write the λmax value for the following compounds.



31. Write the Differences between hard water and soft water.

32. What is the principle of reverse osmosis?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. i) Write the Principle of DTA? (4)
    ii) Differentiate post precipitation from co-pricipitation? (3)

34. i) What is the principle behind sublimation? **(3)**
    ii) Give the uses of sublimation. **(2)**
    iii) Draw a schematic representation of sublimation process? **(2)**

35. i) Explain TGA with an example. **(4)**
    ii) What are the advantages of DME? **(3)**

36. i) Define and explain polarography. **(4)**
    ii) What are the Applications involved polarography? **(3)**

37. i) Write the selection rule of UV spectroscopy. **(3)**
    ii) Explain different types of electronic transitions. **(4)**

38. i) What is temporary hardness? **(2)**
    ii) How to determine the hardness of water? **(3)**
    iii) How to remove the hardness of water? **(2)**

39. i) Write a note on ion-exchange methods. **(4)**
    ii) Advantages and disadvantages of zeolite method. **(3)**

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